10	B	$\frac{GMm}{r^2} = mr\omega^2 \Rightarrow r^3 = \frac{GM}{\omega^2}$ $\Rightarrow r^3 = \frac{(6.67\times10^{-11})(6.0\times10^{24})}{\left(\frac{2\pi}{24\times60\times60}\right)^2} \Rightarrow r = 42297523$ km $v = r\omega = 42297523\times\frac{2\pi}{24\times60\times60} \approx 3080$ m s^{-1}
11	C	Temperature at equilibrium is 273.15 K